

Probability density function *f* (*x*₁, ..., *x*_{*K*−1};*α*₁, ...,*α*_{*K*}) plays a key role in a multifunctional inequality which implies various bounds for the Dirichlet distribution.^[a5]

Related distributions

For *K* independently distributed Gamma distributions:

Y

1

∼
Gamma
(

α

1

,
θ
)
,
.
.
.
,

Y

K

∼
Gamma
(

α

K

,
θ
)

we have:^{[a6]:402}

V
=

∑

i
=
1

K

Y

i

∼
Gamma
(

α

0

,
θ
)
,

X
=
(

X

1

,
.
.
.
,

X

K

)
=
(

Y

1

V

,
.
.
.
,

Y

K

V

)
∼
Dir
(

α

1

,
.
.
.
,

α

K

)
.

Although the *X*_{*s*} are not independent from one another, they can be seen to be generated from a set of *K* independent **gamma** random variable.^{[a6]:594} Unfortunately, since the sum *V* is lost in forming *X* (in fact it can be shown that *V* is stochastically independent of *X*), it is not possible to recover the original gamma random variables from these values alone. Nevertheless, because independent random variables are simpler to work with, this reparametrization can still be useful for proofs about properties of the Dirichlet distribution.

Conjugate prior of the Dirichlet distribution

Because the Dirichlet distribution is an **exponential family distribution** it has a conjugate prior. The conjugate prior is of the form:^[a7]

CD
(
α
|

v
,
η
)
∝
(

1

B
(
α
)

)

η

exp
⁡
(
−

∑

k

v

k

α

k

)
.

Here **v** is a *K*-dimensional real vector and **η** is a scalar parameter. The domain of (**v**,**η**) is restricted to the set of parameters for which the above unnormalized density function can be normalized. The (necessary and sufficient) condition is:^[a8]

∀
k

v

k

>
0

and

η
>
−
1

and

(
η
≤
0

or

∑

k

exp
⁡
−

v

k

η

<
1
)

The conjugation property can be expressed as

if
[
prior
:
α
∼
CD
(
⋅
|

v
,
η
)
]
and
[
observation
:
x
∼
Dirichlet
(
⋅
|
α
)
]
then
[
posterior
:
α
|
x
∼
CD
(
⋅
|

v
+
log
⁡
x
,
η
+
1
)
]
.

In the published literature there is no practical algorithm to efficiently generate samples from CD(**α** | **v**,**η**).

Occurrence and applications

Bayesian models

Dirichlet distributions are most commonly used as the prior distribution of categorical variables or multinomial variables in Bayesian mixture models and other hierarchical Bayesian models. (In many fields, such as in natural language processing, categorical variables are often imprecisely called "multinomial variables". Such a usage is unlikely to cause confusion, just as when Bernoulli distributions and binomial distributions are commonly conflated.)

Inference over hierarchical Bayesian models is often done using Gibbs sampling, and in such a case, instances of the Dirichlet distribution are typically marginalized out of the model by integrating out the Dirichlet random variable. This causes the various categorical variables drawn from the same Dirichlet random variable to become correlated, and the joint distribution over them assumes a Dirichlet-multinomial distribution, conditioned on the hyperparameters of the Dirichlet distribution (the concentration parameters). One of the reasons for doing this is that Gibbs sampling of the Dirichlet-multinomial distribution is extremely easy; see that article for more information.

Intuitive interpretations of the parameters

The concentration parameter

Dirichlet distributions are very often used as prior distributions in Bayesian inference. The simplest and perhaps most common type of Dirichlet prior is the symmetric Dirichlet distribution, where all parameters are equal. This corresponds to the case where you have no prior information to favor one component over any other. As described above, the single value *α* to which all parameters are set is called the **concentration parameter**. If the sample space of the Dirichlet distribution is interpreted as a discrete probability distribution, then intuitively the concentration parameter can be thought of as determining how "concentrated" the probability mass of the Dirichlet distribution to its center, leading to samples with mass dispersed almost equally among all components, i.e., with a value much less than 1, the mass will be highly concentrated in a few components, and all the rest will have almost no mass, and with a value much greater than 1, the mass will be dispersed almost equally among all the components. See the article on the concentration parameter for further discussion.

String cutting

One example use of the Dirichlet distribution is if one wanted to cut strings (each of initial length 1.0) into *K* pieces with different lengths, where each piece had a designated average length, but allowing some variation in the relative sizes of the pieces. Recall that **α**₀ =

∑

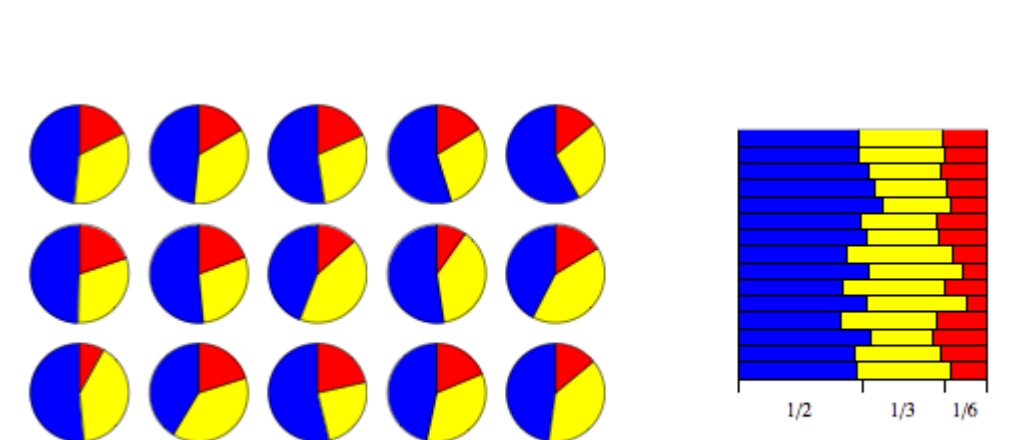
i
=
1

K

α

i

. The *α*_{*i*}/*α*₀ values specify the mean lengths of the cut pieces of string resulting from the distribution. The variance around this mean varies inversely with **α**₀.



Pólya's urn

Consider an urn containing balls of *K* different colors. Initially, the urn contains *a*₁ balls of color 1, *a*₂ balls of color 2, and so on. Now perform *N* draws from the urn, where after each draw, the ball is placed back into the urn with an additional ball of the same color. In the limit as *N* approaches infinity, the proportions of different colored balls in the urn will be distributed as Dir(*a*₁,...,*a*_{*K*}).^[a9]

For a formal proof, note that the proportions of the different colored balls form a bounded [0,1]^{*K*}-valued martingale, hence by the martingale convergence theorem, these proportions converge almost surely and in mean to a limiting random vector. To see that this limiting vector has the above Dirichlet distribution, check that all mixed moments agree.

Each draw from the urn modifies the probability of drawing a ball of any one color from the urn in the future. This modification diminishes with the number of draws, since the relative effect of adding a new ball to the urn diminishes as the urn accumulates increasing numbers of balls.

Random variate generation

From gamma distribution

With a source of Gamma-distributed random variates, one can easily sample a random vector **x** = (*x*₁, ..., *x*_{*K*}) from the *K*-dimensional Dirichlet distribution with parameters (**α**₁, ..., **α**_{*K*}) . First, draw *K* independent random samples **y**₁, ..., **y**_{*K*} from Gamma distributions each with density

Gamma
(

α

i

,
1
)
=

y

i

α

i

−
1

e

−

y

i

Γ
(

α

i

)

,

and then set

x

i

=

y

i

∑

j
=
1

K

y

j

.

[Proof]
The joint distribution of {<i>y</i>_{<i>i</i>}} is given by: $e^{-\sum_i y_i} \prod_{i=1}^K \frac{y_i^{\alpha_i-1}}{\Gamma(\alpha_i)}$ Next, one uses a change of variables, parametrising {<i>y</i>_{<i>i</i>}} in terms of <i>y</i>₁,<i>y</i>₂, ..., <i>y</i>_{<i>K</i>−1} and ∑<!-- ∑ --> i = 1 K y i , and performs a change of variables from <i>y</i> → <i>x</i> such that x K = ∑<!-- ∑ --> i = 1 K y i , <i>x</i>₁ = y 1 x K , <i>x</i>₂ = y 2 x K , ..., <i>x</i>_{<i>K</i>−1} = y K − 1 x K One must then use the change of variables formula. <i>P</i>(<i>x</i>) = <i>P</i>(<i>y</i>(<i>x</i>)) ∂<!-- ∂ --> y ∂<!-- ∂ --> x in which ∂<!-- ∂ --> y ∂<!-- ∂ --> x is the transformation Jacobian. Writing y explicitly as a function of x, one obtains <i>y</i>₁ = <i>x</i>_{<i>K</i>}<i>x</i>₁, <i>y</i>₂ = <i>x</i>_{<i>K</i>}<i>x</i>₂ ... <i>y</i>_{<i>K</i>−1} = <i>x</i>_{<i>K</i>−1}<i>x</i>_{<i>K</i>}, <i>y</i>_{<i>K</i>} = <i>x</i>_{<i>K</i>}(1 − ∑<!-- ∑ --> i = 1 K − 1 x i) The Jacobian now looks like $\begin{vmatrix} x_K & 0 & \cdots & x_1 \\ 0 & x_K & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ -x_K & -x_K & \cdots & 1 - \sum_{i=1}^{K-1} x_i \end{vmatrix}$ The determinant can be evaluated by noting that it remains unchanged if multiples of a row are added to another row, and adding each of the first K-1 rows to the bottom row to obtain $\begin{vmatrix} x_K & 0 & \cdots & x_1 \\ 0 & x_K & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{vmatrix}$ which can be expanded about the bottom row to obtain <i>x</i>_{<i>K</i>}^{<i>K</i>−1} Substituting for x in the joint pdf and including the Jacobian, one obtains: $\frac{\left[\prod_{i=1}^{K-1} (x_i x_K)^{\alpha_i-1}\right] \left[x_K (1 - \sum_{i=1}^{K-1} x_i)\right]^{\alpha_K-1}}{\prod_{i=1}^K \Gamma(\alpha_i)} x_K^{K-1} e^{-x_K}$ Each of the variables 0 ≤ <i>x</i>₁,<i>x</i>₂, ..., <i>x</i>_{<i>k</i>−1} ≤ 1 and likewise 0 ≤ ∑<!-- ∑ --> i = 1 K − 1 x i ≤ 1. Finally, integrate out the extra degree of freedom <i>x</i>_{<i>K</i>} and one obtains: $x_1, x_2, \ldots, x_{K-1} \sim \frac{(1 - \sum_{i=1}^{K-1} x_i)^{\alpha_K-1} \prod_{i=1}^{K-1} x_i^{\alpha_i-1}}{B(\underline{\alpha})}$ Which is equivalent to $\frac{\prod_{i=1}^K x_i^{\alpha_i-1}}{B(\underline{\alpha})} \text{ with support } \sum_{i=1}^K x_i = 1$

Below is example Python code to draw the sample:

```
params = [a1, a2, ..., aK]
xs = (random.betavariate(params[i], sum(params[1:])) for i in range(1, len(params) - 1))
phi = random.betavariate(params[-1], sum(params) - 1)
xs.append((1 - sum(xs)) * phi)
xs.append(1 - sum(xs))
```

This formulation is correct regardless of how the Gamma distributions are parameterized (shape/scale vs. shape/rate) because they are equivalent when scale and rate equal 1.0.

From marginal beta distributions

A less efficient algorithm^[a9] relies on the univariate marginal and conditional distributions being beta and proceeds as follows. Simulate **x**₁ from

Beta
⁡
(

α

1

,

∑

i
=
2

K

α

i

)

Then simulate **x**₂, ..., **x**_{*K*−1} in order, as follows. For *j* = 2, ..., *K* − 1, simulate **φ**_{*j*} from

Beta
⁡
(

α

j

,

∑

i
=
j
+
1

K

α

i

)
,

and let

x

j

=
(
1
−

∑

i
=
1

j
−
1

x

i

)

ϕ

j

.

Finally, set

x

K

=
1
−

∑

i
=
1

K
−
1

x

i

.

This iterative procedure corresponds closely to the "string cutting" intuition described above.

Below is example Python code to draw the sample:

```
params = [a1, a2, ..., aK]
xs = (random.betavariate(params[i], sum(params[1:])) for i in range(1, len(params) - 1))
phi = random.betavariate(params[-1], sum(params) - 1)
xs.append((1 - sum(xs)) * phi)
xs.append(1 - sum(xs))
```

See also

- Generalized Dirichlet distribution
- Grouped Dirichlet distribution
- Inverted Dirichlet distribution
- Latent Dirichlet allocation
- Dirichlet process

- Matrix variate Dirichlet distribution

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External links

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- Dirichlet Distribution (http://users.ics.aalto.fi/ahonkela/dippa/node95.html)
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- SciencesPo (https://cran.r-project.org/web/packages/SciencesPo/index.html): R package that contains functions for simulating parameters of the Dirichlet distribution.

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